

Fresh Produce Warehouse: Picking Performance Analysis

Data Profiling Notes & Insights

1. Dataset Overview

Business context

The data comes from a **fresh fruit and vegetable warehouse** (products such as apples, bananas and tomatoes). Fresh produce warehouses typically work overnight: store orders are picked during the night so that produce can be delivered to stores before they open.

This context matters for every decision in this document:

- **Time pressure:** orders must be ready for early-morning dispatch. Slow picking risks late deliveries and empty shelves.
- **Freshness:** produce has a short shelf life, so stock is usually picked **first-expired-first-out (FEFO)** and may be split across several pallets or batches.
- **Quality:** damaged or unsuitable produce may be rejected or corrected during picking.
- **Seasonality:** the product range changes with the seasons (e.g. new seasonal fruit arriving in spring).
- **Store demand:** stores typically order more before weekends and holidays.

*Note: product IDs are anonymised, so individual products (e.g. which ID is bananas) cannot be identified. Explanations based on the fresh-produce context are **hypotheses** unless the data confirms them.*

The data

The dataset contains **217,963 anonymised picking records** covering 20 days (1–20 April 2023). Each row is one **pick line**: a picker takes a quantity of one product from a storage location and places it into a shipping container (colli) for a store order.

Almost all activity takes place during a **night shift (21:00–03:00)**. The file uses ; as separator and , as decimal, so it must be read with `sep=";"` and `decimal=","`.

Main business question: How can the fresh produce warehouse increase night-shift picking productivity, so that store orders are ready on time, and what factors are currently limiting it?

Purpose of profiling: before cleaning or analysing, check whether each important column is correct, complete and usable. Profiling documents problems; it does not fix them.

How each question is structured:

- **Goal:** why the check is done.
- **What to look for:** what a good or bad result would look like.
- **Findings:** what the notebook output shows.
- **Conclusion:** what it means for cleaning and analysis.
- **Fresh-produce context:** how the finding might be explained in this type of warehouse (hypothesis).

2. Location

Location is the only spatial information in the data and is the basis for any layout analysis (aisle concentration, slotting).

Q1. Is the location format consistent?

Goal: confirm the column can be safely split into its parts.

What to look for:

- Zero missing values.
- Every value following the same ##-##-###-# pattern (0 rows failing).

Findings:

- No missing values and **185 unique locations**.
- Every row follows the same format ##-##-###-# (e.g. 10-06-036-1).

Conclusion: the column is clean and reliable. No cleaning needed.

Q2. Which parts of the location code vary?

Goal: identify which layout levels can be analysed.

What to look for:

- Parts with only one value, which carry no information.
- What each varying part probably represents (aisle, bay, level).

Findings:

- Part 1 (10) and part 4 (1) are constant: likely zone and shelf level.
- Part 2 has **6 values** (aisle) and part 3 has **48 values** (bay).

Conclusion: the layout can only be analysed by **aisle and bay**. Picking height and zones cannot be analysed.

Fresh-produce context: a single zone and a single level fit a produce picking area where crates or pallets stand at floor level, often in one temperature-controlled area. Picking height is therefore not a factor here.

Q3. What is the relationship between location and product?

Goal: know whether location effects can be separated from product effects.

What to look for:

- Whether each product has one location and each location one product (both maximums equal 1).
- Whether any product moved location during the period (a maximum above 1).

Findings:

- 185 products and 185 locations with a **one-to-one mapping**.
- No product changed location during the period (no re-slotting).

Conclusion: location and product are interchangeable. Layout findings must be stated as “location/product”, not blamed on the layout alone.

Fresh-produce context: this matters because produce items differ physically. A location may be slow because it holds heavy or awkward items (e.g. sacks of potatoes, watermelons, boxes of bananas) rather than because of where it is.

Limitation: there are no coordinates or distances, so walking distance cannot be measured.

Think about this before deciding

- Aisles have different numbers of locations. Should aisles be compared by total picks, or by picks per location?
- Because location and product are 1:1, how will you phrase a “slow aisle” finding so it is not blamed on the layout alone (e.g. heavy produce vs poor placement)?
- Without distance data, what layout claims can you honestly make (concentration, time per pick) and which can you not (walking distance)?

3. Volume

Volume represents the physical size of each pick and is a possible measure of workload.

Q1. Is volume complete and stored correctly?

Goal: make sure volume can be used in sums and averages.

What to look for:

- A numeric (float) data type, not text.
- Zero missing values.
- Realistic minimum and maximum values.

Findings:

- Numeric (float) once the comma decimal is read correctly; no missing values.
- Median volume is **0.013**; 75% of picks are 0.022 or less.

Conclusion: complete and usable. The only fix needed was reading the comma decimal correctly.

Fresh-produce context: the unit is not documented. If volume is in **cubic metres**, a median of 0.013 m³ is about 13 litres, roughly one small produce crate, which would be realistic.

Q2. Are there invalid values (zero or negative)?

Goal: find rows that cannot represent a real pick.

What to look for:

- Whether negative rows are concentrated in a few products or collis.
- Whether zero-volume rows have small quantities (rounding) or normal quantities (missing volume).

Findings:

- **5 negative rows**, spread across 3 products (product_83, product_107, product_115). A pick cannot have negative volume, so these are likely **corrections**.
- **199 zero rows** with normal quantities (median 6 units). Their volume is likely **missing rather than rounded**. Most come from a few products (product_111: 20, product_130: 16).

Conclusion: about 0.1% of rows have unreliable volume. Small impact, but each issue needs its own decision.

Fresh-produce context: the negative rows follow very large picks (see Quantity Q3). In produce, this could be an **over-pick being corrected**, or damaged produce being returned during a quality check. Zero volume may mean the product's size was never recorded in the system (missing master data), which is common for new or seasonal items.

Q3. Are there extreme values?

Goal: check whether large values are errors or genuine picks.

What to look for:

- How far the maximum is from the 99th percentile.
- Whether the largest volumes also have large quantities (genuine bulk picks) or not (errors).

Findings:

- Heavily right-skewed: 99% of picks are 0.143 or less, but the maximum is **1.642**. A log-scale chart is needed to see the distribution.
- The largest volumes match large quantities (e.g. 973 units), so they are **genuine bulk picks**.
- 6 of the 10 largest volumes belong to colli_6599.

Conclusion: use the **median, not the mean**, to summarise volume.

Fresh-produce context: a volume above 1 m³ is roughly a full pallet. Full-pallet movements (e.g. a pallet of bananas) are a different kind of work from picking a few crates for a store, which supports excluding them from normal picking metrics.

Q4. Is volume consistent with quantity?

Goal: decide whether volume is a reliable workload measure.

What to look for:

- A strong correlation between volume and quantity.
- For most products, similar minimum, median and maximum volume per unit (small differences are rounding; large ones are problem rows).

Findings:

- Strong correlation between volume and quantity (**r = 0.84**).
- Volume per unit is stable for most products; the 3 products with negative rows show the largest deviations.

Conclusion: volume is a reliable measure of physical workload. Quantity (wm_vso1a) may be enough as the main measure.

Think about this before deciding

- Should the 5 negative rows be **deleted**, or **netted** against the original pick? If they are over-pick corrections, what did the picker actually carry?
- Should the 199 zero-volume rows be kept for pick counts but excluded from volume calculations?
- For produce, is volume a better effort measure than quantity (a crate of watermelons vs a crate of cherry tomatoes)? Or is quantity enough?

4. Colli ID

A colli is the order unit: one shipping container (e.g. a roll cage or crate) for a store order. It is used to measure order size, complexity and completion time.

Q1. Is colli_id complete and consistently formatted?

Goal: make sure every pick can be grouped into an order.

What to look for:

- Zero missing values.
- Every ID following the colli_<number> pattern (0 rows failing).

Findings:

- No missing values and **8,369 unique collis**.
- Every ID follows the colli_<number> format (0 rows fail the pattern).

Conclusion: reliable for grouping picks into orders.

Q2. How many lines does each colli have?

Goal: understand order size, the main measure of order complexity.

What to look for:

- The typical colli size (median).
- How many single-line collis there are.
- Collis far outside the normal range.

Findings:

- Typical colli: **27 lines** (median); 90% have 43 lines or fewer.
- **361 single-line collis** (4%).
- One colli has **727 lines**, far above the 99th percentile of 62.

Conclusion: order sizes vary widely. Single-line collis should be compared separately, and the 727-line colli is a clear outlier.

Fresh-produce context: a typical store order of about 27 different fruit and vegetable lines is realistic for a supermarket's daily produce delivery. Single-line collis could be top-up orders or bulk items (e.g. a full container of one product).

Q3. Is each colli handled by one picker within a realistic time?

Goal: confirm each colli is a real order picked by one person in one session.

What to look for:

- Collis with more than one picker.
- Collis spanning two dates: a short span means a midnight crossing (normal); a long span means a problem.

Findings:

- Only **1 colli** has more than one picker.
- Typical colli takes **7.3 minutes** (median); 99% take under 33 minutes.
- **203 collis span 2 calendar days**, but all except one take 34 minutes or less, so they are midnight crossings.

Conclusion: apart from one colli, all behave like real orders. Crossing midnight is normal for a night shift, so a **shift date** is needed instead of the calendar date.

Q4. What does the suspicious colli look like?

Goal: gather evidence before excluding anything.

What to look for:

- In how many ways the colli breaks the normal pattern (pickers, days, missing values, repeats).
- Whether it is linked to issues found in other columns.

Findings:

- colli_6599 has **727 lines, 36 pickers and 20 days**, with a span of about 19 days (27,617 minutes).
- It also has **9 missing start times** and **581 repeated products**.

Conclusion: not a real order; likely a placeholder or catch-all ID. It is only 0.3% of rows, so **exclude it**.

Fresh-produce context: it also contains the largest quantities (full-pallet volumes). It may record non-order activity such as **stock transfers, replenishment or write-offs of spoiled produce**. This cannot be confirmed from the data.

Q5. Can a colli contain the same product more than once?

Goal: check whether “lines” really means separate picks.

What to look for:

- How many lines repeat a product already in the same colli.
- Whether `wm_tapos` (2 or 3) explains the repeats, or whether they also occur with `wm_tapos = 1`.

Findings:

- **11,723 lines** (5.4% of rows) repeat a product already in the same colli.
- Of these repeated lines, **8,912 (76%)** have `wm_tapos = 2` or `3`, and **2,811 (24%)** have `wm_tapos = 1` (see `wm_tapos` Q3).

Conclusion: `wm_tapos` explains most of the repeats as continuation lines. Raw line counts overstate picks, so decide in cleaning whether a “pick” is a line or a unique product per colli.

Correction: an earlier version compared 13,118 vs 8,912 rows, but the 13,118 included the first line of each repeated product, which is not a repeat.

Think about this before deciding

- Excluding `colli_6599` loses 727 rows (0.3%). Is that acceptable, and how would you justify it to a warehouse manager?
- Should single-line collis be compared with normal store orders when measuring order completion time, or analysed separately?
- Is a repeated line of the same product in one colli **one pick or two**? For example, if a picker takes 5 crates of apples from one pallet and 1 more from the next pallet, is that one pick?

5. Picker

Picker performance is one of the main drivers in the business question. Profiling checks whether each picker has enough reliable data to be compared fairly, not yet who is fast or slow.

Q1. Is picker complete and consistently formatted?

Goal: make sure every pick is linked to one person, with no IDs split or lost.

What to look for:

- Zero missing values.
- Every ID following the `picker_<number>` pattern (0 rows failing).

Findings:

- No missing values and **37 unique pickers**.
- Every ID follows the `picker_<number>` format (0 rows fail the pattern).

Conclusion: the column is clean and reliable. No cleaning needed.

Q2. How many picks and active days does each picker have?

Goal: separate regular pickers from occasional staff, so only like is compared with like.

What to look for:

- Whether the workload is spread evenly or concentrated in a few pickers.
- Whether pickers with few picks also have few active days (occasional staff) or many days with very few picks (not doing regular picking).

Findings:

- Picks per picker range from **1 to 13,837** (median 6,295).
- **28 pickers** have between 1,879 and 13,837 picks; the next picker has only **279**. This is a clear gap between regular and non-regular pickers.
- Among the 28 regular pickers, active days range from **6 to 20**, so some pickers worked only part of the period.
- The 9 low-volume pickers show an unusual pattern: some are active on many days but with very few picks (e.g. picker_31: 15 days, 79 picks; picker_29: 14 days, 279 picks, all in 1 colli).

Conclusion: there are two different groups: **28 regular pickers** and **9 pickers whose activity does not look like normal picking work**. Differences in active days mean totals must be compared per day or per hour, not as raw counts.

Fresh-produce context: produce warehouses often use **temporary or seasonal staff**, which could explain regular pickers with only 6–10 active days. The 9 low-volume users could be supervisors, quality controllers or replenishment staff who occasionally record a pick.

Q3. Are there pickers with too few records to compare fairly?

Goal: set a minimum threshold so that averages are not dominated by a handful of picks.

What to look for:

- A natural gap between regular and occasional pickers.
- How much data would be lost by excluding pickers below a threshold (it should be a small share).

Findings:

- **7 pickers** have fewer than 100 picks (214 picks, 0.10% of rows).
- **9 pickers** have fewer than 500 picks, and the same 9 have fewer than 1,000 (620 picks, **0.28% of rows**).
- The threshold result does not change between 500 and 1,000, which confirms the natural gap between 279 and 1,879 picks.

Conclusion: a threshold of **at least 1,000 picks** keeps all 28 regular pickers and excludes only 0.28% of rows. Exclude the 9 low-volume pickers from picker comparisons.

Q4. Is picker activity linked to the placeholder colli?

Goal: check whether low-volume pickers are real pickers or linked to non-order activity.

What to look for:

- Pickers whose records are mostly or entirely in colli_6599.
- Whether regular pickers are affected by the placeholder colli.

Findings:

- **5 pickers** (picker_29, picker_35, picker_28, picker_36, picker_30) have **100%** of their records in colli_6599.
- **4 more** (picker_31, picker_33, picker_32, picker_34) have **33–75%** of their records there.
- These are the **same 9 low-volume pickers** found in Q3. Regular pickers are barely affected (e.g. picker_17: 13 of 13,837 picks).

Conclusion: the 9 low-volume pickers are closely tied to the placeholder colli, so they are probably not regular pickers. Excluding colli_6599 removes 5 of them entirely. Together with the threshold, the picker analysis should use the **28 regular pickers**.

Think about this before deciding

- What minimum threshold (e.g. at least X picks across Y days) would you use for picker comparisons, and why?
- If some regular pickers are temporary seasonal staff, should they be compared with permanent staff, or is lower speed expected for new workers?
- Should the 9 low-volume users stay in overall totals but be excluded only from picker comparisons?

6. Quantity (wm_vsola)

Quantity is the number of units picked on each line. It is the main measure of workload and will be used for units-per-hour metrics.

Q1. Is wm_vsola complete and stored correctly?

Goal: make sure quantity can be summed and averaged in workload metrics.

What to look for:

- A numeric data type, not text.
- Zero missing values.
- A realistic minimum and maximum.

Findings:

- Numeric (float) once the comma decimal is read correctly; **no missing values** and 165 unique values.
- Quantities range from **1 to 1,008 units**; the median is **6 units** and 75% of lines are 12 units or fewer.

Conclusion: complete and usable. Reading the comma decimal correctly at load time is essential; the old cleaning code turned the whole column into missing values.

Fresh-produce context: the unit is not documented. In a produce warehouse, a unit is most likely a **case or crate** (e.g. a box of apples or a crate of tomatoes), so a typical line of 6 units would be 6 cases for one store.

Q2. Are there invalid values (zero, negative or decimals)?

Goal: find quantities that cannot represent a real pick.

What to look for:

- Zero or negative quantities.
- Non-whole quantities (half units are not possible).

Findings:

- **0 zero, 0 negative** and **0 non-whole** quantities.

Conclusion: every quantity is a valid, whole, positive number. The column can be converted to an integer in cleaning. Unlike volume, quantity has no invalid values, which makes it the more reliable workload measure.

Fresh-produce context: whole numbers support the idea that quantity counts cases or crates rather than weight (kg), which would usually have decimals.

Q3. Are there extreme values?

Goal: check whether large quantities are errors or genuine bulk picks, and whether they would distort averages.

What to look for:

- How far the maximum is from the 99th percentile.
- Whether the largest quantities also have large volumes.
- How many of the largest belong to colli_6599.

Findings:

- Heavily right-skewed: 99% of lines are **56 units or fewer**, but the maximum is **1,008**.
- **9 of the 15 largest quantities** belong to colli_6599, all by picker_35 (a picker with 100% of records in the placeholder colli).

- The other large quantities (384–486 units) are mostly product_83 and product_107 in collis colli_3209, colli_7747, colli_2255 and colli_3901. These are the **same collis that have the negative-volume correction rows**.
- The largest quantities also have large volumes, so they are consistent bulk picks rather than typing errors.

Conclusion: most extreme quantities come from the placeholder colli, which will be excluded. The remaining large picks are linked to corrections, so they should be reviewed together with the negative-volume rows. Use the **median** for typical quantity.

Fresh-produce context: 400+ units on one line is roughly a full pallet of a single product. A pattern of “full pallet picked, then a small correction” could be a pallet sent to a large store with a few damaged cases removed.

Q4. What does quantity look like per colli?

Goal: understand colli workload in units, not just lines.

What to look for:

- How many units a typical colli contains.
- Whether more lines mean more units (a strong correlation), or whether lines and units measure different workload.

Findings:

- A typical colli contains **247 units** (median); 99% contain 482 units or fewer. The maximum (**13,213 units**) is far outside this range.
- A typical line has **6 units** (median per colli), but some collis average up to 180 units per line.
- Lines and units per colli have a **strong but not perfect correlation ($r = 0.79$)**.

Conclusion: lines and units are related but not the same. Two collis with the same number of lines can differ a lot in units. Productivity should be measured with **both lines per hour and units per hour**.

Think about this before deciding

- Should productivity be measured as **lines per hour**, **units per hour**, or both? In produce, picking 10 cases of one product is faster per case than 10 single cases of 10 products. What would a warehouse manager care about more?
- Should extreme quantities (full-pallet lines) be kept, capped or excluded from picking productivity?
- Should the large picks linked to the negative-volume corrections be reviewed together with those correction rows?

7. Timestamps (start_time and finish_time)

Every productivity metric (picks per hour, time per pick, colli duration) is built on these two columns, so this is the most important profiling step.

Q1. Are the timestamps complete, and do the missing values share a pattern?

Goal: know whether missing timestamps are random or linked to specific pickers or collis.

What to look for:

- Which timestamp is complete and which is not.
- Whether missing values belong to regular pickers or to the low-volume pickers.

Findings:

- finish_time is **complete**; start_time has **94 missing values** (0.04%).
- All 94 belong to **4 low-volume pickers** (picker_33, picker_31, picker_32, picker_34) and to collis colli_6599–colli_6608, including the placeholder colli.

Conclusion: the missing start times are not random; they come from the non-regular pickers identified earlier, so they do not affect the regular picking data. finish_time is the more reliable timestamp.

Q2. Do the timestamps fall within the expected period and shift?

Goal: find the actual shift window and how much activity falls outside it.

What to look for:

- The first and last timestamp.
- Where the shift clearly starts and ends.
- How many picks fall outside that window.

Findings:

- Finish times cover **1 April 00:00 to 20 April 23:59**, matching the 20-day period.
- **98.5% of picks** (214,724) finish between **21:00 and 02:59**. The busiest hours are **22:00** (44,505) and **00:00** (43,691).
- Activity drops sharply at 03:00 (2,769 picks), which looks like the end of the shift.
- Only **470 picks** (0.2%) finish between 04:00 and 20:59.

Conclusion: this is a clear **night shift from 21:00 to about 03:00**. Activity outside it is very small and should be checked against the placeholder colli and low-volume pickers in cleaning.

Fresh-produce context: this fits a typical produce operation: produce is picked overnight and loaded around 03:00 for delivery to stores before opening. The sharp drop at 03:00 may be the **dispatch cut-off**. Picks after that could be late orders or re-picks.

Q3. Are the timestamps logically consistent?

Goal: catch system errors with basic logic checks.

What to look for:

- Picks that finish before they start (there should be none).
- Whether calday matches the date of finish_time.

Findings:

- **0 picks** finish before they start.
- calday matches the finish_time date for **every row**.

Conclusion: the timestamps are logically consistent. calday is based on finish_time, so a single night shift is **split across two calday values** at midnight. A shift_date is needed.

Q4. Is start-to-finish a realistic pick duration?

Goal: check whether finish_time – start_time makes physical sense as a pick duration.

What to look for:

- How many picks take 0 seconds.
- How long the longest durations are.

Findings:

- Median start-to-finish duration is **19 seconds**; 95% take 60 seconds or less.
- **9,780 picks (4.5%) take 0 seconds**, and 5% take 1 second or less.
- The longest duration is **2,247 seconds** (about 37 minutes).

Conclusion: a 0-second pick is physically impossible, so start-to-finish is questionable as a pick duration. Q5 checks whether it can be trusted at all.

Q5. Do consecutive picks by the same picker overlap?

Goal: decide whether start-to-finish measures the time a picker actually spends on a pick.

What to look for:

- Whether overlap is rare (a few errors) or the norm (the timestamps mean something else).
- What start_time might actually record, based on an example sequence.

Findings:

- **91.4% of picks start before the picker's previous pick has finished.**
- In the picker_0 example, each new pick starts about 5–10 seconds **before** the previous one is confirmed (e.g. previous finish 00:00:10, next start 00:00:03).
- The first pick in the example even starts on **31 March**, before the data period.

Conclusion: overlap is the norm, not an error. start_time probably records something other than the moment the picker begins the pick (for example, when the task is released or the next location is shown on the scanner). **Start-to-finish should not be used as the pick duration:** summing it would count overlapping time twice.

Q6. What does the time between consecutive picks look like?

Goal: test finish-to-finish as an alternative time metric and find where picking ends and idle time begins.

What to look for:

- Whether a typical gap is realistic for walking to a location and picking.
- The threshold where a gap becomes a break or idle time.
- What 0-second gaps mean.

Findings:

- Median finish-to-finish gap is **12 seconds**; 75% are 20 seconds or less and 90% are 37 seconds or less.
- Long gaps are rare: **5.5% exceed 1 minute**, 1.0% exceed 5 minutes, 0.5% exceed 15 minutes and 0.18% exceed 60 minutes (breaks, end of shift, days off). The maximum is about **13 days**.
- **561 gaps are exactly 0 seconds** (two picks confirmed at the same moment).

Conclusion: finish-to-finish uses only the complete, reliable finish_time and gives realistic values. It is the **strongest candidate for time per pick**, but it needs an **idle threshold** so that breaks and days off are not counted as picking time.

Fresh-produce context: 12 seconds per pick is realistic for walking to the next location and lifting a few cases. Longer gaps may have produce-specific causes: waiting for **replenishment** of an empty location, changing a full roll cage, or a **quality check** of produce. Not all long gaps are breaks.

Think about this before deciding

- Which **idle threshold** separates picking from breaks (1, 5 or 15 minutes)? If a picker waits 3 minutes for a pallet of bananas to be replenished, should that count as picking time?
- Is a 0-second gap (two picks confirmed at the same moment) one pick or two?

- Should the 470 picks after the 03:00 dispatch cut-off be kept, or excluded as non-regular activity?
- Is 91.4% overlap enough evidence to stop using start_time entirely, or is it still useful for anything?

8. Product ID

Product is the “what” of each pick (e.g. apples, bananas, tomatoes, although IDs are anonymised). Because each product has exactly one location (see Location Q3), product demand is also location demand, which makes it the basis for any slotting analysis.

Q1. Is product_id complete and consistently formatted?

Goal: make sure every pick is linked to one product, with no IDs split or lost.

What to look for:

- Zero missing values and zero rows failing the product_<number> pattern.
- Whether the number of products matches the 185 locations.

Findings:

- No missing values and **185 unique products**; every ID follows the product_<number> format.
- 185 products match the 185 locations (1:1, as found in Location Q3).

Conclusion: the column is clean and reliable. No cleaning needed.

Fresh-produce context: 185 products is a realistic fruit and vegetable range for a supermarket supplier.

Q2. How many picks does each product have?

Goal: know how uneven demand is and how many products have enough data to analyse.

What to look for:

- How uneven demand is between products.
- A natural gap between regular and rarely picked products.
- How much data sits in the rarely picked products.

Findings:

- Picks per product range from **1 to 4,269** (median **792**). The top 25% of products have 1,746 picks or more, while the bottom 25% have 390 or fewer.
- **14 products** have fewer than 100 picks, but together they account for only **427 picks (0.20% of rows)**.

- **6 products** have fewer than 10 picks (23 picks, 0.01% of rows).

Conclusion: demand is very uneven across products, which is useful for slotting later. Rarely picked products hold almost no data, so excluding or grouping them has little impact.

Fresh-produce context: this fits produce demand: staples such as bananas, apples, tomatoes and potatoes are ordered by almost every store every day (**fast movers**), while exotic or speciality items are ordered rarely (**slow movers**).

Q3. Are rarely picked products new, discontinued or just slow movers?

Goal: separate data-coverage issues (new or removed products) from genuine low demand.

What to look for:

- Whether rarely picked products appear only near the start or end of the period.
- Whether they are spread across the whole period with few picks (genuine slow movers).

Findings:

- **5 products** (product_177, product_179, product_180, product_181, product_183) first appear on **19 April**, with only 1–2 active days. They look like **new products** introduced at the end of the period.
- **2 products** (product_182, product_184) have a **single pick** on one day (3 and 7 April): one-off picks.
- The other low-volume products (e.g. product_171, product_175, product_178) are picked across **13–17 days** from 1 to 20 April, so they are **genuine slow movers**.

Conclusion: low pick counts have different causes. New products and one-off picks are not comparable with regular products, while genuine slow movers are a real business fact that matters for slotting.

Fresh-produce context: products appearing from 19 April could be **seasonal produce** starting its season in spring (e.g. asparagus or early strawberries in Europe). One-off picks could be promotions or special orders.

Q4. Are any products linked to the placeholder colli?

Goal: check whether excluding colli_6599 would remove or distort any product's data.

What to look for:

- Products whose records are mostly or entirely in colli_6599.
- Whether excluding the placeholder would make any product disappear.

Findings:

- Only **product_184** has all its records in colli_6599, and that is a single pick.
- Every other product has **2% or less** of its picks in the placeholder colli.

Conclusion: excluding colli_6599 has almost no effect on product-level analysis; only one single-pick product disappears.

Think about this before deciding

- Should rarely picked products be **excluded** from product-level comparisons, or **kept** because slow-moving produce (e.g. exotic fruit) matters for slotting?
- Are new seasonal products picked on only 2 days “slow movers”, or just not yet comparable?
- Should new and one-off products be grouped as “other” in the dashboard instead of being removed?

9. wm_tapos

wm_tapos appeared in earlier findings (negative-volume corrections, repeated products in a colli). The wm_ prefix suggests SAP Warehouse Management, where *TAPOS* is usually the **item number within a transfer order**. This section tests that hypothesis with the data.

Q1. What values does wm_tapos take, and how often?

Goal: understand the basic distribution before interpreting it.

What to look for:

- How rare values 2 and 3 are.
- Whether value 3 is a one-off.

Findings:

- **95.9%** of lines have wm_tapos = 1, **4.1% (8,913)** have wm_tapos = 2, and **1 line** has wm_tapos = 3.

Conclusion: almost all lines are 1. Values 2 and 3 matter only for a small, specific group of lines.

Q2. Does wm_tapos = 2 always belong to a product already picked in the same colli?

Goal: test whether value 2 is a continuation of an earlier pick (the SAP item-number hypothesis).

What to look for:

- Whether both shares below are close to 100%.

Findings:

- **99.99%** of $w_m_tapos = 2$ lines have another line of the same product in the same colli.
- **99.98%** of those product groups start with a $w_m_tapos = 1$ line.

Conclusion: the hypothesis is supported: $w_m_tapos = 2$ almost always continues a $w_m_tapos = 1$ pick of the same product in the same colli.

Q3. Do $w_m_tapos = 2$ lines behave differently from normal lines?

Goal: check whether continuation lines represent separate physical work.

What to look for:

- Whether $w_m_tapos = 2$ quantities are much smaller than normal lines.
- How quickly a continuation line follows the previous line of the same product (seconds = same visit; minutes = separate trip).
- How repeated $w_m_tapos = 1$ lines compare.

Findings:

- Continuation lines are small: median **1 unit** (volume 0.003) vs **6 units** (volume 0.014) for normal lines.
- They follow the previous line of the same product within a median of **6 seconds** (75% within 11 seconds, 99% within 55 seconds).
- **2,811** repeated lines have $w_m_tapos = 1$ and follow even faster: median **2 seconds** (75% within 2 seconds).

Conclusion: continuation lines happen during the **same visit** to the location and add only a small remainder. Repeated $w_m_tapos = 1$ lines, confirmed within about 2 seconds, may be **double confirmations**.

Fresh-produce context: this pattern fits **FEFO picking**: the picker takes most cases from the oldest pallet or batch, and when it runs out, takes the remaining 1 case from the next batch. The system records the second batch as a new item line ($w_m_tapos = 2$).

Q4. Are $w_m_tapos = 2$ lines concentrated in specific products, pickers or the placeholder?

Goal: find whether the cause is product-related, picker-related or a data artefact.

What to look for:

- Whether a few products cause most continuation lines.
- Whether the spread across pickers is even (not picker behaviour).

Findings:

- Continuation lines occur across **132 products** and **28 pickers** (the regular pickers); only **11** are in colli_6599.
- They concentrate in a few products: product_65 (383), product_72 (363), product_58 (342), product_76 (300), product_81 (255).
- Across pickers, counts follow workload: the top pickers (picker_7: 605, picker_0: 598) are also among the busiest overall.

Conclusion: the cause is more likely **product-related** than picker behaviour. This could become a business finding later.

Fresh-produce context: products with many split lines may be those whose stock is often spread across **several batches with different expiry dates**, or whose location is too small, so it often runs empty mid-pick. Both are fixable operational issues (replenishment timing, location size).

Q5. What is the single wm_tapos = 3 row?

Goal: check the one unusual value before ignoring it.

What to look for:

- Whether it is a third line of the same product (1 → 2 → 3).

Findings:

- It is the **third line of product_85 in colli_8215**: wm_tapos 1 → 2 → 3 within **5 seconds** (9, 9 and 3 units).

Conclusion: it follows the same continuation pattern as value 2 (e.g. a pick spread across three batches). No special handling needed.

Think about this before deciding

- If wm_tapos = 2 is the same pick continued from the next batch, should it count as **one pick** (merge the lines) or as **two lines** of work? Which makes picks per hour more honest?
- Should the 2,811 repeated wm_tapos = 1 lines confirmed within about 2 seconds be treated as double confirmations and merged?
- If a few products cause most split lines, is that a **business finding** (e.g. replenishment or batch management) worth including in the analysis?

10. Calendar Week (calweek)

calweek is the only column that allows weekly comparisons. With only 20 days of data, the key question is whether the weeks are complete and comparable.

Q1. Is calweek complete and stored correctly?

Goal: basic completeness check and understanding the format (202313 = year 2023, week 13).

What to look for:

- Zero missing values.
- How many weeks there are and how unevenly picks are spread across them.

Findings:

- Integer type with **no missing values**.
- **4 weeks** (202313–202316), with picks per week ranging from **22,652 to 82,197**.

Conclusion: the column is complete, but the large differences in weekly totals need explaining before any weekly comparison (see Q3).

Q2. Is calweek consistent with calday?

Goal: confirm which week rule is used so weekly totals match in SQL and Power BI.

What to look for:

- 0 mismatches between calweek and the ISO week of calday.

Findings:

- **0 rows** where calweek differs from the ISO week of calday.

Conclusion: calweek follows **ISO weeks (Monday to Sunday)** and is derived from calday, so it can be recreated from any date.

Q3. Are all weeks complete?

Goal: check whether weekly totals are comparable.

What to look for:

- Weeks with fewer than 7 days.
- Whether weeks still differ when compared per day instead of in total.

Findings:

- Only **weeks 14 and 15 are complete** (7 days each).
- **Week 13 has only 2 days** (1–2 April, Saturday–Sunday) and **week 16 has only 4 days** (17–20 April, Monday–Thursday).
- Per day, weeks 13, 15 and 16 are similar (**11,326–11,742 picks/day**), while **week 14 is lower (9,457 picks/day)**.

Conclusion: weekly **totals are misleading** because weeks cover different numbers of days. Compare **picks per day**. The lower week 14 is worth investigating at daily level in EDA.

Fresh-produce context: week 14 (3–9 April 2023) includes **Good Friday (7 April) and Easter weekend**. In many countries, stores close or change opening hours over Easter, which would reduce produce orders. This is a hypothesis to check at daily level; the data does not state the country.

Q4. Is every weekday equally represented?

Goal: make sure weekday comparisons are not biased by how often each weekday occurs.

What to look for:

- Whether each weekday occurs the same number of times.
- The effect of the midnight split on weekday assignment.

Findings:

- **Friday occurs only twice**; every other weekday occurs 3 times.
- Friday has the **fewest total picks (24,608)**, but per occurrence it has about **12,300 picks**, the highest of all weekdays. Thursday is the lowest per occurrence (about 9,000).
- calday is based on finish_time, so picks after midnight are assigned to the **next weekday**. For example, “Friday” picks mix the end of Thursday’s night shift with the start of Friday’s.

Conclusion: weekday totals are misleading: Friday only looks quiet because it occurs less often. Weekday analysis must use **averages per occurrence** and a **shift date**, not calday.

Fresh-produce context: high Friday volume fits stores **stocking up on fresh produce before the weekend**. Of the only two Fridays in the data, one is Good Friday, so this pattern needs care.

Correction (see D26): with shift_date, there are only two full Friday shifts and one is Good Friday (15,450 picks), so the Friday result comes from Easter and the midnight split, not from a weekly pattern.

Think about this before deciding

- With only 2 complete weeks (one of them including Easter), is weekly analysis meaningful, or should the analysis use **daily** (per shift_date) figures instead?
- Should Easter days be flagged as **holidays** so they do not distort normal-day comparisons?
- Should calweek be kept, dropped, or recalculated from shift_date? In Power BI, would a **date table** (shift date, week, weekday, holiday flag) be better?

11. Data Quality Log

Column	Issue	Rows affected	Suggested decision
All	Comma used as decimal separator	All rows	Read with decimal=", " at load time
area, warehouse, wm_vltyp, calmonth, calyear	Only one value	All rows	Drop
wm_srsrc, wm_qdocno	100% / 99.99% empty	All rows	Drop
location	No coordinates or distances	—	Note as a limitation
volume	Negative values (volume corrections on wm_tapos = 2 lines)	5	Decided (D4): net into the original pick
volume	Zero values (missing volume; 192 in colli_6599)	199	Decided (D5): keep rows, set volume to NaN
colli_id	colli_6599 is a placeholder (non-order activity)	727	Decided (D7): exclude
colli_id	Collis crossing midnight	203 collis	Keep; create shift_date
colli_id	Single-line collis	361 collis	Decided (D8): keep, flag, exclude from colli-level metrics
colli_id	Same product repeated in a colli	11,723	Decided (D9): one pick per product, colli and visit
wm_tapos	Continuation lines (wm_tapos = 2/3, likely next batch)	8,912 (8,903 after D7, D10)	Decided (D23): merge into the original pick; flag has_continuation
wm_tapos	Repeated wm_tapos = 1 lines within about 2 seconds	2,811 (2,239 excl. placeholder)	Decided (D9): merge, sum quantities
picker	9 low-volume users linked to colli_6599	620 (85 after D7)	Decided (D10, D12): threshold 1,000 picks; keep in totals,

Column	Issue	Rows affected	Suggested decision
			exclude from time metrics
picker	Shifts worked differ between pickers (4–18 shifts; “6–20 days” was inflated by the midnight split)	—	Decided (D11): core vs occasional segment; compare rates, not totals
wm_vsola	Stored as float, but all values are whole numbers	All rows	Convert to integer
wm_vsola	Extreme quantities (up to 1,008, full-pallet size)	Mostly colli_6599; 563 lines of 100+ cases after D7	Decided (D7, D14, D15): placeholder excluded; keep and flag bulk lines; corrections netted (D4)
start_time	Missing values (all from 4 low-volume pickers)	94	Removed with the low-volume pickers / placeholder
start_time	Next pick starts before previous finishes (overlap)	91.4% of picks	Decided (D19): do not use start_time for any metric
start_time	Zero-second start-to-finish durations	9,780	Resolved by D19: start-to-finish not used
finish_time	Activity outside the 21:00–03:00 shift	470 (04:00–20:59)	Resolved by D7 and D18: 466 in colli_6599; 4 real picks kept
finish_time	Idle gaps between picks (breaks, replenishment, days off)	2,236 gaps > 5 min	Decided (D16): gaps of 15 min or more are breaks
finish_time	Zero-second gaps (simultaneous confirmations)	561 (100 after D7, D10)	Decided (D17): two picks (different collis)
calday	Night shift split at midnight	All rows	Decided (D28): create shift_date
product_id	New (possibly seasonal) products	5 products	Decided (D21): flag as new; not

Column	Issue	Rows affected	Suggested decision
product_id	from 19 April One-off products (single pick)	2 products (1 after D7)	compared Decided (D20): below the 100-pick threshold for time comparisons
product_id	Rarely picked products (< 100 picks)	14 products (13 after D7), 0.20% of rows	Decided (D20, D22): keep; 100-pick threshold for time comparisons; “Other” in visuals
calweek	Partial weeks (week 13: 2 days, week 16: 4 days)	69,569	Decided (D26): shift-level analysis; no weekly trend claims
calweek / calday	Friday occurs twice, other weekdays three times	—	Decided (D26): no weekday trend claims (Friday result driven by Good Friday)
calweek / calday	Easter inside the period (low and peak shifts 5–10 April)	6 shifts	Decided (D27): flag as holiday period; analyse separately
calweek / calday	Weekday and week based on finish date (midnight split)	All rows	Decided (D28): drop; derive from shift_date
shift_date	First (31 Mar) and last (20 Apr) shifts are partial	2 shifts	Decided (D29): flag; exclude from shift-level comparisons

12. Decisions Log

Each decision records **what** was decided, **why** (with evidence), a **concrete example**, the **alternatives that were rejected**, and the **impact** on the next steps. Decisions are made after profiling and before cleaning.

Summary

ID	Topic	Decision
D1	Location	Compare aisles with three measures: total picks, picks per location and time per

ID	Topic	Decision
		pick
D2	Location	Report slow aisles as location/product effects, controlling for quantity
D3	Location	Only make layout claims the data supports; add a Limitations section
D4	Volume	Net the 5 negative-volume rows into their original pick
D5	Volume	Keep zero-volume rows; set their volume to missing (NaN)
D6	Volume	Output: cases primary, volume secondary. Speed: lines per hour primary, cases per hour secondary (updated with D13)
D13	Quantity	Use lines per hour as the primary productivity KPI and cases per hour as the secondary output KPI
D14	Quantity	Keep bulk lines (100+ cases) without capping; flag them and run a sensitivity check
D15	Quantity	Large picks linked to corrections are covered by D4; no extra action
D16	Timestamps	Measure time with finish-to-finish gaps within a shift; gaps of 15 minutes or more are breaks
D17	Timestamps	Zero-second gaps in different collis count as two picks (follows D9)
D18	Timestamps	Keep the 4 remaining afternoon picks; the idle threshold handles their gap
D19	Timestamps	Do not use start_time for any metric; keep it for reference

ID	Topic	Decision
		only
D20	Product ID	Keep all products; compare time per pick only for products with at least 100 picks
D21	Product ID	Flag new products; do not compare them with regular products
D22	Product ID	Show top products plus an “Other” group in dashboard visuals
D23	wm_tapos	Merge continuation lines into the original pick; keep the last line’s finish time; flag the split
D24	wm_tapos	Include split lines by product as an EDA question and candidate business insight
D25	wm_tapos	Fast repeated wm_tapos = 1 lines are covered by D9
D26	Calendar	Use the shift (shift_date) as the main time unit; no weekly or weekday trend claims
D27	Calendar	Flag shifts 5–10 April as a holiday period; analyse them separately
D28	Calendar	Drop calweek and calday; derive week, weekday and holiday flag from shift_date (date table in Power BI)
D29	Calendar	Flag the partial first and last shifts; exclude them from shift-level comparisons
D7	Colli ID	Exclude the placeholder colli_6599 from all picking analysis

ID	Topic	Decision
D8	Colli ID	Keep single-line collis as picks, but flag them and exclude them from colli-level metrics
D9	Colli ID	One pick = one product into one colli during one visit; merge fast repeats and sum their quantities
D10	Picker	Compare only regular pickers with at least 1,000 picks
D11	Picker	Keep pickers with few shifts, but segment pickers into core and occasional; compare rates, not totals
D12	Picker	Keep the 4 remaining low-volume users in volume totals, but exclude them from time and productivity metrics

D1. Compare aisles with three measures

Decision: compare aisles using **total picks** (workload and congestion), **picks per location** (demand intensity and slotting) and **time per pick** (performance). Each measure is used for its own purpose.

Why:

- Aisles have very different sizes: from **16 locations** (aisle 04) to **41** (aisle 06). A large aisle will always have more total picks simply because it has more products, so total picks alone mixes up *size* and *demand*.
- Each measure answers a different business question for a different person:
 - **Total picks:** “Where is the workload, and where might pickers get in each other’s way?” (shift manager)
 - **Picks per location:** “Which aisles hold the fast-moving produce?” (layout planner)
 - **Time per pick:** “Where is picking slow?” (the productivity question itself)

Example (from initial exploration; to be confirmed in EDA):

Aisle	Total picks	Locations	Picks per location
-------	-------------	-----------	--------------------

Aisle	Total picks	Locations	Picks per location
02	13,881	32	~430
04	10,276	16	~640

On total picks, aisle 02 looks **busier** than aisle 04. Per location, aisle 04 is **about 50% busier**: its products are ordered more often. Using totals alone would lead to the wrong conclusion about where the fast movers are.

Alternatives rejected:

- *Total picks only*: hides demand intensity and makes large aisles look important just because they are large.
- *Picks per location only*: hides where the actual workload and possible congestion are.

Impact: EDA and SQL calculate all three measures per aisle. The Power BI layout page shows them side by side.

D2. Report slow aisles as location/product effects, controlling for quantity

Decision: never state that an aisle is slow *because of the layout*. Report slow aisles as **location/product** effects, keep **observations separate from possible explanations**, and **control for quantity** (compare within quantity bands or per case) before comparing aisles. Also check continuation lines (`wm_tapos = 2`) and picker mix.

Why:

- Each location holds exactly **one product**, and no product moved during the period (Location Q3). So a slow location could be slow because of **where** it is or because of **what** is stored there, and the data cannot fully separate the two.
- In fresh produce, products differ a lot physically: a sack of potatoes or a watermelon is heavier and bulkier than a punnet of cherry tomatoes, and fragile fruit (berries) must be handled carefully.
- Quantity is the most obvious measurable cause of slow picks: picking 20 cases takes longer than picking 1. Without controlling for it, an aisle full of large-quantity picks will always look slow.

Example:

-  *Overclaiming*: “Aisle 05 has a poor layout and slows pickers down.”
-  *Honest*: “Picks in aisle 05 take longer than average. Each location holds one product, so this may reflect the produce stored there rather than the aisle position. The difference remains when comparing only picks of 1–5 cases, so it is not explained by quantity.”

Alternatives rejected:

- *Compare raw time per pick by aisle:* would blame aisles for large orders or bulky produce.
- *Claim a layout effect:* not supported, because location and product cannot be separated.

Impact: EDA compares aisles within quantity bands (e.g. 1–5, 6–12, 13+ cases) and uses volume per case as a size control (see D6). All insights use careful wording.

D3. Only make layout claims the data supports

Decision: make only these layout claims: **concentration (Pareto)**, **time per pick by aisle and bay**, **congestion** (pickers in the same aisle at the same time), **aisle changes per colli** (as a proxy for travel) and **split lines by product**. Do **not** claim walking distance, travel time vs handling time, an “optimal” layout or savings in metres. Add a **Limitations** section to the final report.

Why:

- The data has **no coordinates or distances**. We do not know how far aisle 01 is from aisle 06, or where the dispatch area is.
- Finish-to-finish time includes **both walking and handling**, and the two cannot be separated.
- Counting how often a picker switches aisles within one order is the closest honest proxy for travel: more switches suggest more walking, even if the distance is unknown.

Example:

✓ Can say	✗ Cannot say
“20% of locations account for X% of picks.”	“Moving fast movers closer to dispatch would save 2 km per shift.”
“Orders that switch aisles more often take longer per pick.”	“Pickers spend 40% of their time walking.”
“Aisles 01 and 06 often have several pickers working at the same time.”	“Aisle 06 is congested because it is too narrow.”

Alternatives rejected:

- *Estimate distances from location codes* (e.g. assume aisles are 5 m apart): the numbers would look precise but be invented.
- *Skip layout analysis entirely:* would lose useful, supported insights (concentration, congestion, aisle switching).

Impact: SQL calculates aisle changes per colli with window functions (LAG). The final report includes a Limitations section.

D4. Net the 5 negative-volume rows into their original pick

Decision: merge each negative-volume row into its original pick: **add the quantities and add the volumes**, so each pair becomes **one pick**.

Why:

- Each negative row is a **wm_tapos = 2 continuation line** of a very large pick of the **same product in the same colli**. The wm_tapos analysis showed that wm_tapos = 2 lines continue the original pick, so they belong to it.
- The negative row's **quantity is positive** (1–8 cases). Only its **volume** is negative, so it is a **volume correction**, not a return of produce.
- After netting, the volume matches the **expected volume** (total cases × the product's normal volume per case) in **all 5 cases**.

Example (colli_7747, product_107):

	Cases	Volume
Line 1 (wm_tapos = 1)	424	0.945
Line 2 (wm_tapos = 2)	1	−0.201
Netted (one pick)	425	0.744
Expected (425 × normal volume per case)		0.744

The first line **overstates** the volume by about 27%; the second line **corrects** it.

Alternatives rejected:

- *Delete the 5 rows:* would lose 1–8 real cases per pick **and** leave the large picks' volume overstated by about 25%. Deleting removes the symptom but keeps the error.
- *Set the negative volume to 0:* same problem; the overstated volume on line 1 would remain.

Impact: in cleaning, these 5 pairs are merged into 5 picks. The same netting logic applies to the wider “what counts as one pick” decision for wm_tapos = 2 lines.

D5. Keep zero-volume rows; set their volume to missing (NaN)

Decision: keep the rows with zero volume, but set their volume to **missing (NaN)**.

Why:

- **192 of the 199** zero-volume rows are in colli_6599, which is excluded anyway (placeholder). Only **7 rows** remain.

- These rows have **normal quantities** (median 6 cases), so they are real picks; only the volume is missing (probably not recorded in the product master data).
- A zero is a false value: it says “this pick had no size”, which is impossible. Missing (NaN) honestly says “we don’t know”.

Example: a picker picks 6 cases of a product whose volume was not recorded.

Option	Pick count	Average volume
Keep volume = 0	✓ Counted	✗ Pulled down by a false 0
Delete the row	✗ One real pick lost	✓ Not affected
Set volume = NaN	✓ Counted	✓ Not affected (NaN is ignored in averages)

Alternatives rejected:

- *Delete the rows:* loses real picks from productivity metrics.
- *Estimate the volume* (cases × normal volume per case): adds made-up values for only 7 rows, with no real benefit.

Impact: pick counts stay complete; volume averages in pandas, SQL (AVG ignores NULL) and Power BI are not distorted.

D6. Workload and speed measures (updated with D13)

Decision:

- **Workload / output:** use **quantity (cases)** as the primary measure and **volume** as secondary. Volume is used to classify products by size (e.g. small, medium, bulky by volume per case) and as a control when comparing aisles and products.
- **Productivity / speed:** use **lines per hour** as the primary KPI and **cases per hour** as secondary (see D13).
- Keep volume KPIs off the main dashboard page.

Update: the first version of D6 said “cases per hour” was the primary workload KPI. This was refined in D13, because picking time depends mostly on the visit, not on the number of cases.

Why:

- **Quantity is the more reliable column:** complete, no zero, negative or decimal values. Volume has invalid values and an undocumented unit (probably m³).
- **Cases are a standard warehouse output measure** that managers understand immediately. “Cubic metres” is harder to explain.

- But in fresh produce, **one case is not always the same amount of work**. Volume captures physical effort that quantity misses, which is exactly the control needed for D2.

Example: two pickers each pick **100 cases per hour**.

- Picker A picks mostly **cherry tomatoes** (small, light cases).
- Picker B picks mostly **watermelons and potatoes** (bulky, heavy cases).

On cases alone, they look equally productive. But picker B handled far more physical volume. Without volume as a secondary measure, picker B could be unfairly ranked as average, or an aisle with bulky produce could be wrongly labelled slow.

Alternatives rejected:

- *Volume as the primary measure:* less reliable, unit unknown, and harder for managers to understand.
- *Drop volume entirely:* would lose the only way to account for product size, weakening fair comparisons of pickers and aisles.

Impact: output KPIs use cases; speed KPIs use lines per hour (D13). EDA creates a product size class from volume per case, used to compare like with like.

D7. Exclude the placeholder colli_6599 from all picking analysis

Decision: exclude colli_6599 (727 rows, 0.3%) from all picking and productivity analysis, and document it as a known limitation.

Why: colli_6599 breaks every pattern of a real store order:

Check	colli_6599	A normal colli
Lines	727	27 (median)
Pickers	36	1
Days	20 (span of about 19 days)	1 shift
Zero-volume rows	192 of the 199 in the dataset	almost none
Largest quantities	9 of the top 15 (up to 1,008 cases)	median 6
Off-shift activity (04:00–20:59)	466 of the 470 off-shift picks	almost none

It is not a store order. It most likely records **non-order activity** (e.g. stock transfers, replenishment or write-offs of spoiled produce) under one catch-all ID. This cannot be confirmed from the data.

Example (how to explain it to a warehouse manager): “One container ID contains 727 lines from 36 pickers over 19 days, mostly outside the night shift. That is not a store order;

it looks like other warehouse activity recorded under one ID. It is only 0.3% of rows, so leaving it out does not change the overall picture, while keeping it would distort order sizes, colli times and quantities.”

Keeping it would make one “order” 27 times larger than normal and 19 days long, which would distort every colli average.

Alternatives rejected:

- *Keep it in the analysis:* distorts colli size, colli duration, quantity extremes and off-shift activity.
- *Keep it as a separate “non-order activity” category:* more complete, but its content cannot be explained, so it adds little value to the productivity question.

Impact: removes 727 rows, and with them 466 of the 470 off-shift picks, 192 zero-volume rows, 9 missing start times and most extreme quantities. 5 of the 9 low-volume pickers disappear completely.

D8. Keep single-line collis, but flag them and exclude them from colli-level metrics

Decision: keep the 361 single-line collis as real picks, add a flag (e.g. `is_single_line = True`), and **exclude them from colli-level metrics** (colli duration, lines per colli). Include them in pick-level metrics.

Why:

- They look like **real small store orders** (e.g. top-ups), not errors: median **6 cases** (the same as a normal line), picked during normal shift hours, across 81 products and 32 pickers.
- But a single-line colli has only one pick, so its colli duration (first to last pick) is always **0 seconds**, which makes average colli time look faster than it is.

Example: a picker completes 10 store orders of 27 lines in about 7 minutes each, plus 2 single-line orders.

Average colli duration	
Normal orders only	7.0 minutes
Including single-line orders: $(10 \times 7 + 2 \times 0) \div 12$	5.8 minutes , which looks faster although nothing got faster
Include single-line collis?	
Metric	
Picks per hour, time per pick (finish-to-finish)	Yes: each pick is real work

Metric	Include single-line collis?
Colli duration, lines per colli	No: separate them, or they distort the averages

Alternatives rejected:

- *Delete single-line collis*: loses real picks and real store orders.
- *Treat them like normal orders everywhere*: makes colli duration look shorter than it really is.

Impact: cleaning adds an `is_single_line` flag. Colli-level queries in SQL and Power BI measures filter on this flag.

D9. One pick = one product into one colli during one visit; merge fast repeats

Decision: define **one pick = one product picked into one colli during one visit**. Merge the fast repeated `wm_tapos = 1` lines of the same product in the same colli into **one pick**, and **sum their quantities**. State the assumption.

Why:

- Excluding the placeholder, there are **2,239** repeated `wm_tapos = 1` lines. **96.7%** are confirmed within **2 seconds** of the previous line of the same product.
- In 2 seconds a picker cannot walk to another location and back, so these lines come from the **same visit**.
- Two explanations are possible, and the data cannot tell which is true:

Explanation	What happened	Cases
A. Split confirmation	The picker scans 3 cases, then 3 more from the same location	Both real, so sum them
B. Double scan	The picker scans the same 3 cases twice by mistake	Only 3 real, so summing overstates

- **62%** have the same quantity as the previous line, which points somewhat towards B, but does not prove it.
- Both explanations agree it is **one visit**, so it should count as **one pick**. They only disagree on the quantity. Summing treats what the system recorded as the truth, because double scans cannot be verified.
- This is the **same definition** used in D4, so the meaning of “one pick” stays consistent.

Example: in `colli_1031`, `product_158` is confirmed with 9 cases, and again with 9 cases 2 seconds later.

Approach	Picks	Cases
Count every line	2	18
One pick, summed quantity (chosen)	1	18
One pick, maximum quantity	1	9

Counting every line would credit the picker with two picks for one visit, which inflates picks per hour.

Alternatives rejected:

- *Count every line as a pick*: inflates picks per hour by counting one visit twice.
- *Merge and take the maximum quantity*: assumes every repeat is a double scan, which would understate genuine split confirmations.

Impact: affects 1.0% of lines and at most **0.5% of cases** (the maximum possible overstatement if all repeats were double scans). This impact is small and is documented as an assumption. The same definition is applied to the `wm_tapos = 2` continuation lines (confirmed in D23).

D10. Compare only regular pickers with at least 1,000 picks

Decision: a picker is included in picker comparisons only with **at least 1,000 picks**.

Why:

- After excluding `colli_6599` (D7), only **4 low-volume users** remain below the threshold, with **6–36 picks** each. The next picker has **1,879 picks**. Nothing lies between 36 and 1,879, so this is a **natural gap** in the data, not an arbitrary cut-off. Any threshold from about 100 to 1,800 gives the same result.
- The 4 users also differ in **kind**, not just in volume:

User	Picks	Shifts	Missing start_time	Typical lines per colli
picker_33	36	4	36 (100%)	1
picker_32	23	9	23 (100%)	1
picker_31	20	11	20 (100%)	1
picker_34	6	4	6 (100%)	1

- They have **no start times at all**, work almost only on **single-line collis**, and make about **2–9 picks per shift** compared with hundreds for regular pickers. They are probably not order pickers (e.g. supervisors or staff doing exception or manual picks).

Example: picker_31 made 20 picks over 11 shifts, about 2 per shift. An average time per pick based on 2 picks per shift would be dominated by the hours between them, so a comparison with a picker making 700 picks per shift would be meaningless.

Alternatives rejected:

- *A lower threshold (e.g. 100 picks):* gives the same result, but looks arbitrary and sits close to the low group.
- *A role-based rule only* (“has start times and multi-line collis”): more precise, but harder to explain in one line. It is used as supporting evidence instead.

Impact: picker comparisons use the **28 regular pickers**. The threshold is simple to apply in SQL and Power BI.

D11. Keep pickers with few shifts, but segment them; compare rates, not totals

Decision: keep all 28 regular pickers. Add a **segment** based on shifts worked: **core** (9 or more shifts, 21 pickers) and **occasional** (6 or fewer shifts, 7 pickers). Compare pickers using **rates** (e.g. picks per hour), never totals, and show **distributions** (median and quartiles) rather than one ranked list.

Why:

- **Shifts, not calendar days, measure time worked.** Counting days with calday split each night shift at midnight, so one shift often counted as two days. Counting shifts, regular pickers worked **4–18 shifts** (not “6–20 days”).
- **7 pickers worked only 4–6 shifts** (picker_27, 14, 22, 13, 26, 19 and 20), but each still has **1,879–3,970 picks**, which is enough data for reliable averages. Excluding them is not necessary.
- No regular picker worked 7 or 8 shifts, so the segment boundary also follows a **natural gap**.
- In fresh produce, **temporary and seasonal staff** are common. A new worker in their 4th shift does not yet know the layout and the products. Ranking them against someone in their 18th shift compares **experience**, not ability.

Example: if a manager sees a ranking in which the bottom five are all occasional pickers, the wrong conclusion is “these people are slow; don’t rehire them”. The useful question is “do new workers get faster with experience, and how quickly?” Segmenting makes that question answerable.

Alternatives rejected:

- *Exclude pickers with few shifts:* loses 7 pickers with plenty of data, and a possible business insight about onboarding.
- *One combined ranking:* unfair, because it mixes experience with performance.
- *Compare total picks:* rewards pickers for working more shifts, not for picking faster.

Impact: cleaning creates shift_date and a picker_segment column. EDA tests the hypothesis “**Do pickers get faster over their first shifts?**”, which could lead to a

recommendation about training temporary staff. The segment boundaries are a first proposal to confirm in EDA.

D12. Keep the 4 remaining low-volume users in totals, but exclude them from time metrics

Decision: keep the rows of the 4 remaining low-volume users and flag them (e.g. `is_regular_picker = False`). Include them in **volume totals** (picks, cases shipped), and exclude them from **all time and productivity metrics** and picker comparisons.

Why:

- Their **85 picks (0.04% of rows)** are real work sent to stores, so removing them would slightly understate the warehouse’s total output.
- But their time data is unusable: they have **no start times**, and with 2–9 picks per shift, the gaps between their picks are hours long, which says nothing about picking speed.

Metric	Include?	Reason
Total picks and cases shipped	Yes	Real work sent to stores
Time per pick, picks per hour	No	Gaps between picks are hours long; no start times
Picker comparisons	No	Below the D10 threshold

Example: picker_34 made 6 picks over 4 shifts. Their “time per pick” would be several hours, which would pull the warehouse’s average time per pick up and make the whole operation look slower than it is.

Alternatives rejected:

- *Delete their rows:* loses real output from volume totals.
- *Include them everywhere:* distorts time metrics with meaningless gaps.

Impact: the same **flag, don’t delete** pattern as D8. SQL queries and Power BI measures filter time metrics on `is_regular_picker = True`.

D13. Lines per hour is the primary productivity KPI; cases per hour is secondary

Decision: measure picking **speed** with **lines per hour** (primary) and picking **output** with **cases per hour** (secondary). When comparing pickers or aisles, control for the order mix (quantity bands, as in D2).

Why:

- The time a pick takes depends mostly on the **visit** (walking to the location and scanning), not on the number of cases:

Cases on the line	Lines	Median time per pick (finish-to-finish)	Time per case
1	35,417	7 s	7.0 s
2–5	61,419	12 s	~3.4 s
6–12	78,006	13 s	~1.4 s
13–56	40,395	16 s	~0.6 s
57+	1,967	58 s	~0.6 s

- Going from 1 case to 6–12 cases barely doubles the time. Each extra case adds only about a second; the fixed cost is the visit. One line = one visit, so lines per hour measures the work the picker actually controls.
- The number of cases per line is decided by the **store order**, not by the picker. Cases per hour therefore depends on the order mix a picker happened to receive.

Example:

- Picker A's orders need **10 lines of 1 case**: 10 visits, about 70 seconds, **10 cases**.
- Picker B's order needs **1 line of 60 cases**: 1 visit, about 58 seconds, **60 cases**.

On cases per hour, picker B looks about 6 times more productive, although neither picker chose their orders. Ranking pickers by cases per hour would reward the order mix, not skill.

Measure	Best use	Who uses it
Lines per hour	Picker productivity and speed	Shift manager, picker comparisons
Cases per hour	Capacity: how many pickers are needed for the volume	Planning

Alternatives rejected:

- *Cases per hour only*: rewards pickers who receive large-quantity orders.
- *Lines per hour only*: loses the output view needed for capacity planning.

Impact: the main productivity KPI in SQL and Power BI is lines per hour (using the pick definition in D9); cases per hour is shown as a secondary output KPI.

D14. Keep bulk lines without capping; flag them and run a sensitivity check

Decision: keep all large-quantity lines as they are (no capping, no exclusion). Add a flag for **bulk lines (100 or more cases)** and, in EDA, check whether conclusions change when bulk lines are excluded.

Why:

- Most extreme quantities were in the placeholder colli and are removed by D7. After D7:

Threshold	Lines	Share of lines	Share of cases
100+ cases	563	0.26%	4.1%
200+ cases	63	0.03%	0.9%
400+ cases	3	0.001%	0.07%

- These are **real store picks** (e.g. colli_3209, product_83, 486 cases), not errors.
- Their effect on **lines per hour** (D13) is tiny: 563 lines out of about 217,000. Median-based measures are not pulled by extremes. They only matter noticeably for **cases per hour**.

Example: capping a 486-case line at 100 cases would make the total cases in the analysis no longer match what was actually shipped to the store, and would understate output by up to 4% of cases.

Alternatives rejected:

- *Exclude bulk lines:* loses real store orders and understates output.
- *Cap bulk lines:* changes real data, so totals no longer match what was shipped.

Impact: cleaning adds an `is_bulk_line` flag (100+ cases). EDA reports whether the main conclusions hold with and without bulk lines.

D15. Large picks linked to corrections are covered by D4

Decision: no extra action. The large picks with negative-volume corrections are handled by D4 (netting).

Why:

- The 5 large picks with corrections (e.g. colli_7747, 424 + 1 cases) are merged with their correction line in D4, so their cases and volume are already correct.
- Most of the remaining bulk lines (300–486 cases) are product_83 and product_107. This is noted for EDA, not treated as a data problem.

Alternatives rejected:

- *Treat the large picks as errors and remove them:* they are real store picks with consistent quantity and volume.

Impact: none beyond D4. product_83 and product_107 are noted for the EDA product analysis.

D16. Finish-to-finish time within a shift, with a 15-minute idle threshold

Decision: measure time per pick as the **finish-to-finish gap** between a picker's consecutive picks **within the same shift**. Gaps **under 15 minutes** count as working time (including order changeovers and short waits). Gaps of **15 minutes or more** are breaks and are excluded.

Why:

- Measured after D7 and for regular pickers only (D10, D12), gaps longer than 2 minutes form **three groups**:

Gap length	Number of gaps	What they look like
2–10 min	about 6,600, falling steadily	95% happen between two different collis: changeovers from one store order to the next (new roll cage, new labels)
10–16 min	only about 60	A clear trough
16–30 min	about 630, peaking at about 20 minutes	Concentrated at 23:00, 01:00 and 03:00 , about 2 per picker per shift : scheduled breaks

- The threshold is placed **in the trough**, between the changeovers and the break cluster, so it follows the data rather than an arbitrary number.
- Changeovers are **real work**: finishing an order and starting the next one is part of picking.
- Waiting for replenishment cannot be separated from other pauses in the data. Short waits count as working time, which reflects the picking process as a manager experiences it.
- Gaps are measured **within a shift**, so the time between one shift and the next never counts.

Example: how each threshold changes the result:

Threshold	Median time per pick	Working hours counted	Effect
5 min	12 s	1,220 h	Removes about 126 h of order changeovers (real work), so productivity looks better than it is
10 min	12 s	1,336 h	Close to the trough,

Threshold	Median time per pick	Working hours counted	Effect
			but slightly inside the tail of the changeovers
15 min	12 s	1,346 h	In the trough, just before the break cluster

The median stays at 12 seconds with every threshold. The threshold mainly changes **total working hours**, and therefore **picks per hour**.

Alternatives rejected:

- *5-minute threshold*: treats order changeovers as breaks and overstates productivity.
- *No threshold*: counts scheduled breaks as picking time and understates productivity.
- *Start-to-finish time*: unreliable (see D19).

Impact: time per pick and picks per hour in cleaning, SQL and Power BI use finish-to-finish gaps within a shift, excluding gaps of 15 minutes or more.

D17. Zero-second gaps in different collis count as two picks

Decision: when two picks are confirmed at the same second, count them as **two picks**. No special rule is needed.

Why:

- After D7 and D10, only **100 zero-second gaps** remain (down from 561).
- **None** of them are in the same colli; 61% are the same product.
- They are therefore **two different store orders** confirmed at the same moment (e.g. picking the same product into two store cages side by side).
- Under the pick definition in D9 (one product into one colli during one visit), different collis mean **two picks**.

Example: a picker places cases of the same product into two store cages and confirms both at the same second. These are two store orders, so two picks.

Alternatives rejected:

- *Merge them into one pick*: would undercount real work for two different stores.

Impact: none beyond applying D9 consistently.

D18. Keep the 4 remaining afternoon picks

Decision: keep the 4 picks outside the night shift that remain after D7. Include them in totals. No extra rule is needed for time metrics.

Why:

- D7 already removed **466 of the 470** off-shift picks (they were in colli_6599).
- The remaining **4 picks** are one real store order (colli_7683), picked by picker_6 on 14 April between 15:37 and 15:49.
- The gap between this order and the evening shift is longer than 15 minutes, so D16 excludes it from time metrics automatically.

Alternatives rejected:

- *Exclude them:* would remove a real store order for no benefit.

Impact: none beyond D7 and D16.

D19. Do not use start_time for any metric

Decision: do not use start_time in any metric. Keep the column in the cleaned data for reference only, and explain in the report why it was not used.

Why:

- **91.4%** of picks start before the picker's previous pick has finished (Timestamps Q5).
- **4.5%** of picks have a start-to-finish time of 0 seconds (Timestamps Q4).
- All **94 missing values** come from the placeholder colli and the low-volume users (Timestamps Q1).
- Even for the **first pick of a shift**, the start is a median of **33 seconds** before the finish, longer than a normal pick (12 seconds), so it does not reliably mark when work began.
- finish_time is complete and covers everything needed: time per pick, shift start and end, and colli duration.

Example: the first version of this project measured pick duration as start-to-finish. Because 91.4% of picks overlap, adding those durations counts the same time twice. This is why the project switched to finish-to-finish time (D16).

Alternatives rejected:

- *Keep start-to-finish as the pick duration:* double-counts overlapping time and includes impossible 0-second picks.
- *Use start_time only for the first pick of each shift:* it does not reliably mark the start of work either.

Impact: all time metrics use finish_time only. The report explains the change from the first version.

D20. Keep all products; compare time only for products with at least 100 picks

Decision: keep **all products** in totals and in the demand/slotting analysis. Include only products with **at least 100 picks** in **product-level time comparisons** (e.g. time per pick by product).

Why:

- After D7, **184 products** remain (product_184 disappears with the placeholder). **13 products** have fewer than 100 picks, together only **0.20% of picks**.
- With very few picks, a product's average time is unreliable: one slow or fast pick decides the result.
- But slow-moving produce still **occupies a location**. For slotting, it matters which locations hold slow movers, because a slow mover in a prime location wastes that location. Excluding them would hide this.
- This is the same principle as **D10** for pickers: enough data before comparing.

Group	Picks	Median time per pick
Regular products (100+ picks)	171 products	typically 10–20 s (median 14 s)
Slow movers (e.g. product_171, product_175, product_178, product_174, product_103)	33–92 picks over 11–16 shifts	12–21 s, mostly within the normal range
New products and one-offs	1–11 picks	21–209 s, too few picks to be reliable

Example: product_182 has **1 pick**, and its “median time per pick” is **209 seconds**. That is a single pick, not a pattern. Ranking it as the slowest product would be misleading.

Alternatives rejected:

- *Exclude rarely picked products everywhere:* loses slow movers from the slotting analysis.
- *Include them everywhere:* gives unreliable time figures for 13 products.

Impact: product-level time comparisons in EDA, SQL and Power BI filter on products with 100+ picks. Totals and demand analysis include all products.

D21. Flag new products; do not compare them with regular products

Decision: flag the new products (e.g. `is_new_product`) and do not compare them with regular products. Mention them in the insights.

Why:

- **5 products** (`product_177`, `product_179`, `product_180`, `product_181`, `product_183`) first appear in the **last 2 shifts** (18–20 April), with only **2–11 picks** each.
- Their median times (21–40 s) look slower than regular products (14 s), but with so few picks it is not possible to tell whether this is real or noise.
- The 100-pick threshold in D20 already excludes them from time comparisons, so no extra rule is needed.

Example: `product_181` has 4 picks over 2 shifts. Comparing its time per pick with a product picked thousands of times over 20 days would not be a fair comparison.

Alternatives rejected:

- *Compare them with regular products:* too few picks for a reliable comparison.
- *Remove them:* loses real picks, and the fact that new products were introduced at the end of the period.

Impact: cleaning adds an `is_new_product` flag. The insights note that 5 products were introduced at the end of the period, so a future analysis with more data can check them.

D22. Show top products plus an “Other” group in dashboard visuals

Decision: in product visuals, show the **top products** (e.g. the top 20 by picks) and group the rest as “**Other**”. The exact number is decided in the Power BI stage.

Why:

- A product chart with 184 bars is unreadable.
- Removing products from visuals would make the totals disagree with what was shipped. Grouping keeps totals complete.
- This is standard dashboard practice.

Example: a bar chart of picks by product shows 20 named products and one “Other” bar. The total of all bars still equals the total picks.

Alternatives rejected:

- *Show all 184 products:* unreadable.
- *Show only the top products without “Other”:* totals no longer match.

Impact: Power BI product visuals use a top-N + “Other” grouping.

D23. Merge continuation lines into the original pick

Decision: merge each continuation line ($w_{m_tapos} = 2$ or 3) into the original pick of the same product in the same colli. **Add up** the cases and the volume (as in D4). The merged pick keeps the **finish time of its last line**. Add a flag (e.g. $has_continuation = True$).

Why:

- After D7 and D10, **8,903 continuation lines** remain (4.1% of lines).
- A continuation line is confirmed a median of **6 seconds** after the first line (99% within **54 seconds**), so it happens during the **same visit** to the location.
- Its median quantity is **1 case**: the remainder of the pick (w_{m_tapos} Q3).
- Under the pick definition in **D9** (one product into one colli during one visit), a continuation line is part of the original pick. This is the same logic as **D4**.
- Keeping the **last line's finish time** means the extra seconds spent on the continuation stay inside that pick's time. Nothing is lost; it is just not counted as a separate pick.

Example: a picker picks 6 cases of a product into a store cage ($w_{m_tapos} = 1$), then 1 more case 6 seconds later ($w_{m_tapos} = 2$).

Approach	Picks	Cases	Time counted
Count every line	2	7	split over 2 picks
Merge (chosen)	1	7	the full time, including the extra 6 seconds

Counting every line would inflate picks per hour by about **4%**, crediting the picker with a second “pick” for a 6-second remainder at the same location.

Alternatives rejected:

- *Count continuation lines as separate picks:* inflates picks per hour by about 4%.
- *Delete continuation lines:* loses real cases and the time spent on them.

Impact: cleaning merges continuation lines into one pick with summed cases and volume, the last finish time, and a $has_continuation$ flag used in D24.

D24. Include split lines by product as a candidate business insight

Decision: include **split lines by product** (how often a product needs a continuation line) as an **EDA question and candidate business insight**. Keep what the data shows separate from possible explanations, as in D2 and D3.

Why:

- Most products rarely need a second line, but a **small group needs one in 40–72% of picks** (products with 100+ picks, following D20):

Product	Picks	Continuation lines	Split rate
product_72	502	363	72%
product_65	697	382	55%
product_172	103	50	49%
product_81	527	254	48%
product_88	468	224	48%

All products **4.3%** overall; median product 3%

- The **top 10 products** account for **31% of all continuation lines**.
- Continuation lines add about **23 hours** of picking time over 20 days, roughly **1.7%** of total working hours (1,346 h, D16).
- It is concentrated, measurable and points to a process the warehouse could change, so it connects directly to the business question: *what is limiting productivity?*

The data shows	Possible explanations (cannot be confirmed)
Some products need a second line in 40–72% of picks	Stock split across batches with different expiry dates (FEFO)
This adds about 23 hours of picking time in 20 days	The location is too small, so it runs empty mid-pick
It is concentrated in about 10 products	Replenishment is not keeping up during the shift

Example of an honest insight: “10 products require a second pick line in 40–72% of picks, adding about 23 hours of picking time over 20 days. This is consistent with stock being split across batches or locations running empty mid-pick. Reviewing replenishment timing and location capacity for these products could reduce this.”

Alternatives rejected:

- *Treat split lines only as a data issue:* would miss a measurable, actionable finding.
- *State a cause (e.g. “replenishment is too slow”):* not supported by the data.

Impact: EDA calculates the split rate and extra time per product using the `has_continuation` flag (D23). SQL and Power BI can show the products with the highest split rates.

D25. Fast repeated `wm_tapos = 1` lines are covered by D9

Decision: no new decision. The fast repeated `wm_tapos = 1` lines are merged and their quantities summed, as decided in **D9**.

Why: D9 already defines one pick as one product into one colli during one visit, and covers these lines (2,239 after D7, 96.7% within 2 seconds).

Impact: none beyond D9.

Picks per shift (reference for D26–D29)

Picks per shift, using shift_date (the evening a shift starts), with colli_6599 excluded:

Shift (evening of)	Weekday	Picks	Pickers
31 Mar	Fri	6,244 (partial)	21
1 Apr	Sat	11,876	18
2 Apr	Sun	9,640	18
3 Apr	Mon	10,985	17
4 Apr	Tue	12,608	21
5 Apr	Wed	5,485 (low)	12
6 Apr	Thu	5,367 (low)	9
7 Apr	Fri (Good Friday)	15,450 (peak)	25
8 Apr	Sat	5,790 (low)	12
9 Apr	Sun (Easter Sunday)	6,467 (low)	16
10 Apr	Mon (Easter Monday)	16,744 (peak)	25
11–19 Apr	—	10,409–12,547	14–21
20 Apr	Thu	6,595 (partial)	15

D26. Use the shift as the main time unit; no weekly or weekday trend claims

Decision: use the **shift (shift_date)** as the main time unit for all analysis. Show weekly figures only as context, and make **no weekly or weekday trend claims**. Note this in the Limitations section.

Why:

- Only **2 complete weeks** exist, and one of them contains Easter.
- **Week 14 is not “a slow week”**: it is a normal Monday–Tuesday followed by Easter dips *and* peaks. A weekly average hides this.
- **Correction to the calendar-week profiling (Calendar Week Q4)**: with calday, Friday looked like the busiest weekday. With shift_date, there are only two full Friday shifts, and one is **Good Friday (15,450 picks)**. The result came from Easter and the midnight split, not from a weekly pattern.
- 6 of the 19 full shifts are in the holiday period (D27), which leaves only **1–2 normal shifts per weekday**. That is too few to claim patterns such as “Mondays are busier”.

Example: a weekly chart would show week 14 as the lowest week (about 8,900 picks per shift) and suggest a slowdown. The shift-level table shows it is actually two low days, two peak days and two more low days around Easter.

Alternatives rejected:

- *Weekly analysis:* too few complete weeks, and one is distorted by Easter.
- *Weekday comparisons:* too few normal shifts per weekday.

Impact: EDA, SQL and Power BI use `shift_date` as the main time dimension. The report's Limitations section states that the 20-day period is too short for weekly or weekday patterns.

D27. Flag shifts 5–10 April as a holiday period; analyse them separately

Decision: flag shifts **5–10 April** as `holiday_period = True`. Base normal-day comparisons on the other shifts, and analyse the holiday period **separately** (how productivity and staffing behave on low and peak shifts).

Why:

- The data shows an unusual pattern, not just a dip:
 - **Low shifts** (5,367–6,467 picks, about half of normal) on 5, 6, 8 and 9 April.
 - **Peak shifts** (15,450 and 16,744 picks, about 40% above normal) on 7 and 10 April.
 - Staffing follows the same pattern: **9–12 pickers** on low shifts and **25** on peak shifts.
- This is **consistent with Easter** (Good Friday 7 April, Easter Sunday 9 April, Easter Monday 10 April 2023): stores order less before closed days and catch up when they reopen. The country is not stated, so this is not confirmed.
- **Keeping it in normal averages** would distort them, because the lows and peaks do not cancel out cleanly.
- **Removing it** would lose the only **peak-volume shifts** in the data. How productivity holds up at 16,744 picks compared with about 11,000 is valuable for the productivity question.

Example: the average picks per shift over all full shifts mixes 5,367-pick and 16,744-pick shifts with normal ones. Separating the holiday period shows the normal level (about 10,400–12,600 picks per shift) and the peaks separately.

Alternatives rejected:

- *Treat Easter shifts as normal:* distorts normal-day averages.
- *Remove Easter shifts:* loses the only peak-volume shifts.

Impact: cleaning adds a holiday_period flag. EDA compares productivity and staffing on normal, low and peak shifts.

D28. Drop calweek and calday; derive calendar fields from shift_date

Decision: drop calweek and calday. Create shift_date in cleaning, and derive **week, weekday and the holiday flag** from it. In Power BI, build a **date table** keyed on shift_date.

Why:

- calday is based on finish_time, so every night shift is **split across two dates** at midnight (Timestamps Q3).
- calweek is derived from calday (Calendar Week Q2), so it has the same split. For example, the after-midnight part of Sunday's shift falls into **the next week**.
- A date table is standard star-schema practice in Power BI and keeps all calendar fields consistent.

Example: Sunday 9 April's shift runs from 21:04 to 00:58. With calday, its picks after midnight are counted on Monday 10 April, which is a different day **and** a different ISO week.

Alternatives rejected:

- *Keep calweek and calday:* every day and week total would mix two different shifts.
- *Keep both calday and shift_date:* risks using the wrong one by mistake.

Impact: cleaning creates shift_date and drops calday and calweek. The Power BI date table is designed in the Power BI stage.

D29. Flag the partial first and last shifts

Decision: flag the **31 March** and **20 April** shifts as partial_shift = True. Exclude them from **shift-level comparisons** (e.g. picks per shift). Keep them in **pick-level metrics** (time per pick, picks per hour).

Why:

- The data starts at **1 April 00:00** and ends at **20 April 23:59**.
- The **31 March shift** only includes its after-midnight part (6,244 picks), and the **20 April shift** only its before-midnight part (6,595 picks).
- Both look like “low days”, but only because half of each shift is missing.
- Each individual pick in these shifts is still valid, so pick-level metrics can use them.

Example: in a chart of picks per shift, 20 April would appear as one of the lowest shifts (6,595 picks), although it stops at midnight, in the middle of the shift.

Alternatives rejected:

- *Treat them as full shifts:* makes them look like low-volume shifts.

- *Delete them*: loses valid picks for pick-level metrics.

Impact: cleaning adds a `partial_shift` flag. Shift-level queries and visuals filter on complete shifts (19 shifts).

All profiling decisions are complete (D1–D29). Next step: apply them in data cleaning.

13. Open Questions

These are the main decisions to make before cleaning, in order of impact:

1. ~~What counts as one pick?~~ **Decided (D9, D23):** one product into one colli during one visit, including continuation lines.
2. ~~Time metric and idle threshold~~ **Decided (D16, D19):** finish-to-finish within a shift; gaps of 15 minutes or more are breaks; `start_time` not used.
3. **Exclusions Decided:** placeholder colli (D7), off-shift picks (D7, D18), low-volume users (D10, D12), rare and new products (D20, D21).
4. ~~Corrections: delete or net the 5 negative volume rows?~~ **Decided (D4): net.**
5. ~~Productivity measure: lines per hour, cases per hour, or both?~~ **Decided (D13): lines per hour primary, cases per hour secondary.**
6. ~~Shift date and holidays~~ **Decided (D26–D29):** shift-level analysis; holiday period flagged; `calweek/calday` replaced by `shift_date`; partial shifts flagged.

Profiling complete. Next step: decide these questions before cleaning.